

Human-Computer Interaction A5 (2025/2026)

Team Name

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1 Introduction

This report presents the final stage of the **EcoGo** project, a **Smart Sustainable Travel Companion** designed to guide users toward eco-friendly travel choices. Starting from the Needfinding phase (A1) and proceeding through Low-Fi (A2) and Mid-Fi (A4) prototyping, we have now developed an interactive High-Fidelity prototype.

The goal of this assignment is to validate our design decisions through the implementation of a functional software prototype and to assess its usability through testing with real users. The evaluation focuses on the three core tasks defined in previous assignments: discovering eco-friendly options, planning a safe trip, and tracking sustainable progress.

2 High-Fidelity Prototype

2.1 Prototype Overview

The High-Fidelity (Hi-Fi) prototype is a responsive **Web Application**. It builds upon the grayscale wireframes from Assignment 4 by adding color, interactivity, and real data persistence. While the Mid-Fi prototype focused on layout and flow, this version simulates the final “look and feel” of the product, allowing users to experience the application as if it were a live service.

2.2 Tools, Frameworks, and Technologies

To ensure a scalable and realistic user experience, we selected the following technology stack:

- **Frontend Framework (Vue 3 + Vite):** We chose Vue.js for its reactivity and component-based architecture, allowing us to build complex interfaces like the Dashboard and Map view efficiently. Vite ensures rapid development and loading speeds.
- **Styling (TailwindCSS + DaisyUI + Lucide.vue + Turf.js):** Used to implement a consistent, responsive visual design with pre-built accessible components (cards, buttons, navbars, routes) that adhere to modern UI standards.
- **Backend & Database (Node.js, Express, MongoDB):** Unlike simple local prototypes, our system connects to a real MongoDB Atlas database via a Node/Express API. This allows for genuine user authentication, data storage, and retrieval.
- **External APIs:**
 - **Mapbox GL JS, Mapbox Directions, Mapbox search:** For the interactive map functionalities, allowing users to visualize routes and terrain.
 - **Google Gemini AI:** Integrated to generate dynamic, context-aware sustainable travel tips, comparing and calculating co2 emissions, help to find out alternative travel methods.

2.3 Prototype Functionality

The prototype implements the core user journey for the three defined tasks, supported by real-time and AI features:

- **Task 1: Discover Eco-Friendly Options (Simple).** Users can access the “Tips” section, where the backend calls the Google Gemini AI to generate unique daily sustainability advice. Additionally, the “Live” section monitors tourist crowding and suggests alternative, less crowded destinations.

- **Task 2: Plan a Trip (Moderate).** The "Plan" page integrates Mapbox to allow users to visualize itineraries. The system provides a comparison of emissions, costs, and time between different transport modes (Train vs. Car), helping users make informed green choices.
- **Task 3: Track & Reflect (Complex).** The "Dashboard" and "Rewards" pages are fully interactive. They display the user's "Eco-Points" and "CO₂ Saved" based on their travel history. This section also includes Gamification elements (leaderboards, badges) to value sustainable choices.

Limitations: While the routing visualization is interactive, the actual booking payment is simulated (no real banking connection), and real-time weather alerts are based on mocked data streams for demonstration purposes, also there is no real tracking of traveling, so user just manually complete when he arrived on destination.

2.4 Persistence and Data Handling

We implemented a robust persistence strategy using **MongoDB Atlas** and **Mongoose**.

- **User Data:** Profiles, including roles (Standard, Admin, Admin for Forum), email, and hashed passwords, are stored in the `users` collection.
- **Content:** User-generated content, such as Travel Cards and Feedback messages, are stored in respective collections (`TravelCard`, `Feedback`), ensuring that community contributions persist across sessions.
- **Session Management:** We use `localStorage` to store JSON Web Tokens (JWT) for maintaining user login sessions and UI preferences (Dark/Light mode).

2.5 Visual Design

The visual design reinforces the theme of sustainability through a **Nature-Inspired Palette**, utilizing shades of Emerald and Forest Green (`bg-emerald-50`, `text-green-700`) to evoke calmness and eco-consciousness. The layout follows a **Card-Based UI** pattern (using DaisyUI cards), which effectively organizes complex data—such as trip details or leaderboard stats—into digestible chunks. Typography is handled by sans-serif fonts for maximum readability on both desktop and mobile devices.

2.6 Prototype Access

The source code and instructions are available in our repository:

- **GitHub Repository:** <https://github.com/schwein69/Asw-Hci>

3 Usability Test Methodology

3.1 Participants

We recruited **2 participants** to conduct the usability testing. To ensure authenticity and relevance, both participants are real users from our target demographic (University students) .

- **Yanik Fatula (Unibo Student, 23):** Represents our “Immediate User” persona. A Computer Science student who travels frequently by train and is highly tech-savvy.
- **Isa Yaqoubi (Unibo Student, 26):** Represents the “Extreme User” profile. He is a working student interested in sustainability but typically prioritizes speed; he matches the target audience for our gamification features.
- **Farinelli Simone (University of Bristol Student, 24):** Represents the “Immediate user” profile. He is a student interested in Game develop and design. He lives abroad, always on travelling and is comfortable using technology.

3.2 Roles

Explain the role of each team member during the usability tests.

- **Facilitator:** Kalile Horri (Guided the user through the session, managed consent forms).
- **Note-taker:** Jiahao Guo (Recorded errors, quotes, and non-verbal cues).
- **Observer(s)** Intisar bel meddeb (Identified critical incidents and usability bottlenecks that might be missed during note-taking)

3.3 Tasks

The participants were asked to perform three tasks corresponding to the prototype’s core features:

1. **Discover (AI & Tips):** “Find a daily sustainable travel tip generated by the system and view the current crowding status of your city.”
2. **Plan (Map & Booking):** “Plan a trip from Paris to Berlin using the map interface. Select the option with the lowest CO₂ emissions.”
3. **Track (Gamification):** “Check your current Eco-Points on the leaderboard and view your environmental impact statistics.”

3.4 Procedure

The test followed the **Think Aloud** protocol.

1. **Introduction:** We explained the purpose of EcoGo and obtained informed consent.
2. **Execution:** Users performed the 3 tasks while vocalizing their thoughts.
3. **Debriefing:** We administered the Single Ease Question (SEQ) and asked open-ended questions about the “Live” notifications and AI features.

3.5 Materials

- Hi-fi prototype (running on localhost)
- Laptop for testing + Smartphone for recording audio
- Consent forms (allowed by participants)

3.6 Ethics and Consent

All participants allowed an informed consent form prior to the test. They explicitly agreed to the use of their real names and data for the purpose of this academic assignment verification.

4 Usability Test Results

4.1 Observations

- **Positive:** P1 was visibly impressed by the AI-generated tips, stating, *"we didn't expect it to give us actual new advice every time."*
- **Positive:** P1 appreciated the **Dark Mode** toggle, noting that it made the map interface much more comfortable to use during the simulated "nighttime" planning task.
- **Positive:** P2 found the real-time alerts in the Notification section very useful, remarking, *"It's great that it notifies me about traffic disruptions and weather alerts, these are very useful"*
- **Ambiguity:** P2 was momentarily confused by the "Live" icon in the navigation bar.
- **Positive:** P3 expressed a strong preference for the Discover feature, stating, *"I absolutely love this part; finding hidden sustainable spots is usually so hard, but this makes it exciting."*
- **Negative:** P3 was confused by the Gamification metrics, asking, *"I see the Eco-Points on the leaderboard, but I don't understand how they actually translate to my real-world impact."*

4.2 Task Performance

Task	Success	Time	Key Issue
1. Discover (AI)	100%	30s	None.
2. Plan (Map)	100%	95s	Zooming on map felt sensitive.
3. Track (Stats)	100%	45s	None.

Table 1: Task Performance Metrics

Task	Success	Time	Key Issue
1. Discover (AI)	100%	60s	None.
2. Plan (Map)	100%	50s	None.
3. Track (Stats)	100%	100s	Not very impressed by stats.

Table 2: Task Performance Metrics

Task	Success	Time	Key Issue
1. Discover (AI)	100%	10s	None.
2. Plan (Map)	100%	95s	Lagging.
3. Track (Stats)	50%	90s	First time didn't found dashboard.

Table 3: Task Performance Metrics

5 Discussion and Proposed Improvements

The usability testing sessions revealed that the **EcoGo** prototype effectively communicates its core value proposition: making sustainable travel choices visible and rewarding.

What worked well:

- **AI Integration:** The dynamic generation of tips via Google Gemini was a standout feature. Participants felt the content was fresh and relevant, preventing the "banner blindness" often seen with static advice.
- **Gamification:** The "Eco-Points" and Leaderboard created an immediate sense of accomplishment. Participant P2 specifically mentioned that seeing their rank made them want to book the greener option.

What did not work:

- **System Feedback Latency:** When fetching AI tips, the interface simply "froze" for a second before displaying text. Without a loading indicator, P1 wasn't sure if the button click had registered.
- **Dashboard not very clear:** P2 and P3 were struggled to interpret the environmental impact graph, stating, *"I see the numbers, but I don't intuitively understand if my CO2 savings are actually significant compared to a standard trip."*

5.1 Identified Issues

1. **Loading States (H1):** The AI generation taking 1-2 seconds sometimes looked like the page was unresponsive before the text appeared.

5.2 Proposed Changes

- **Floating Feedback Action:** Add a persistent "Feedback" bubble in the bottom-right corner to encourage user reporting as per the project requirements.
- **Restyling dashboard:** Optimize dashboard's visualization.

6 Conclusion

The development of the Hi-Fi prototype successfully validated the integration of advanced features like AI and Maps into a user-friendly interface. The MEVN stack provided the necessary robustness for data persistence, while the usability tests confirmed that users value the transparency of emissions data and the motivation provided by the gamification elements. Future work will focus on refining the real-time notification system and smoothing out map interactions.